

Newman-Penrose Formalism Calculations

Stephani et al., Equation (13.51)

Metric

$$ds^2 = \left(-x_4^{\frac{2a_2^2}{a_2+a_3} + \frac{2a_2a_3}{a_2+a_3} + \frac{2a_3^2}{a_2+a_3} - \frac{2a_2}{a_2+a_3} - \frac{2a_3}{a_2+a_3}} \right) dx_4 \otimes dx_4 + x_4^{2a_3} dx_3 \otimes dx_3 + x_4^{2a_2} dx_2 \otimes dx_2 + \left(\frac{1}{x_4^{\frac{2a_2a_3}{a_2+a_3}}} \right) dx_1 \otimes dx_1$$

Coordinates

$$(x^\mu) = (x_4, x_3, x_2, x_1)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} x_4^{a_2 - \frac{a_2a_3}{a_2+a_3} + a_3 - 1} \right) dx_4 + \left(\frac{1}{2} \sqrt{2} x_4^{a_3} \right) dx_3$$

$$n = \left(\frac{1}{2} \sqrt{2} x_4^{a_2 - \frac{a_2a_3}{a_2+a_3} + a_3 - 1} \right) dx_4 + \left(-\frac{1}{2} \sqrt{2} x_4^{a_3} \right) dx_3$$

$$m = \left(\frac{1}{2} \sqrt{2} x_4^{a_2} \right) dx_2 + \left(\frac{i \sqrt{2}}{2 x_4^{\frac{a_2a_3}{a_2+a_3}}} \right) dx_1$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} x_4^{a_2} \right) dx_2 + \left(-\frac{i \sqrt{2}}{2 x_4^{\frac{a_2a_3}{a_2+a_3}}} \right) dx_1$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2 a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3} + \frac{a_2}{a_2+a_3} + \frac{a_3}{a_2+a_3}} \right) dx_4 + \left(\frac{\sqrt{2}}{2 x_4^{a_3}} \right) dx_3$$

$$n = \left(-\frac{1}{2} \sqrt{2} x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2 a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3} + \frac{a_2}{a_2+a_3} + \frac{a_3}{a_2+a_3}} \right) dx_4 + \left(-\frac{\sqrt{2}}{2 x_4^{a_3}} \right) dx_3$$

$$m = \left(\frac{\sqrt{2}}{2 x_4^{a_2}} \right) dx_2 + \left(\frac{1}{2} i \sqrt{2} x_4^{\frac{a_2 a_3}{a_2+a_3}} \right) dx_1$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2 x_4^{a_2}} \right) dx_2 + \left(-\frac{1}{2} i \sqrt{2} x_4^{\frac{a_2 a_3}{a_2+a_3}} \right) dx_1$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2 a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3} + \frac{a_2}{a_2+a_3} + \frac{a_3}{a_2+a_3}} \partial_{x_4} + \frac{\sqrt{2}}{2 x_4^{a_3}} \partial_{x_3}$$

$$\Delta = -\frac{1}{2} \sqrt{2} x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2 a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3} + \frac{a_2}{a_2+a_3} + \frac{a_3}{a_2+a_3}} \partial_{x_4} - \frac{\sqrt{2}}{2 x_4^{a_3}} \partial_{x_3}$$

$$\delta = \frac{\sqrt{2}}{2 x_4^{a_2}} \partial_{x_2} + \frac{1}{2} i \sqrt{2} x_4^{\frac{a_2 a_3}{a_2+a_3}} \partial_{x_1}$$

$$\bar{\delta} = \frac{\sqrt{2}}{2 x_4^{a_2}} \partial_{x_2} - \frac{1}{2} i \sqrt{2} x_4^{\frac{a_2 a_3}{a_2+a_3}} \partial_{x_1}$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} a_2^2 x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2 a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3}}}{4 (a_2 + a_3)}$$

$$\sigma = \frac{\sqrt{2}(a_2 + 2a_3)a_2x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3}}}{4(a_2 + a_3)}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}a_2^2x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3}}}{4(a_2 + a_3)}$$

$$\lambda = -\frac{\sqrt{2}(a_2 + 2a_3)a_2x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3}}}{4(a_2 + a_3)}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{1}{4}\sqrt{2}a_3x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3}}$$

$$\gamma = \frac{1}{4}\sqrt{2}a_3x_4^{-\frac{a_2^2}{a_2+a_3} - \frac{a_2a_3}{a_2+a_3} - \frac{a_3^2}{a_2+a_3}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = \frac{(a_2 + 2a_3)a_2a_3x_4 - \frac{2a_2^2}{a_2+a_3} - \frac{2a_2a_3}{a_2+a_3} - \frac{2a_3^2}{a_2+a_3}}{2(a_2 + a_3)}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{a_2^2a_3x_4 - \frac{2a_2^2}{a_2+a_3} - \frac{2a_2a_3}{a_2+a_3} - \frac{2a_3^2}{a_2+a_3}}{2(a_2 + a_3)}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{(a_2 + 2\,a_3)a_2a_3x_4-\frac{2\,a_2^2}{a_2+a_3}-\frac{2\,a_2a_3}{a_2+a_3}-\frac{2\,a_3^2}{a_2+a_3}}{2\,(a_2+a_3)}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"